# HU Collection

TMUA Mock Set D Paper 2

Worked Solutions

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**Answer key**

| Q | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans | C | B | B | D | E | C | F | D | C | A |
| Q | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans | D | A | D | B | C | A | B | E | D | E |

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

**Question 1. Answer C.**

The statement has three layers: for every class, there exists a student, such that for every problem, the student solved it. Negating swaps each quantifier and negates the innermost claim: there is a class such that for every student there is a problem the student did not solve. That is option C. Option F negates only the middle layer, and option D keeps "every class" unchanged.

**Question 2. Answer B.**

A counterexample must make the hypothesis true and the conclusion false. For $x^3$ at 0 the gradient is zero but the curve keeps increasing through the origin: a point of inflection, not a turning point. Options A, C, D and F have $f'(a) = 0$ and a genuine turning point, and option E has $f'(0) = 1 \neq 0$, so its hypothesis is false and it cannot be a counterexample.

**Question 3. Answer B.**

If $4 \mid n$ then $4 \mid n(n+3) = n^2 + 3n$, so the condition is sufficient. It is not necessary: $n = 1$ gives $n^2 + 3n = 4$, divisible by 4, although $4 \nmid 1$. In general $n(n+3)$ is divisible by 4 when $n \equiv 0$ or $1 \pmod 4$.

**Question 4. Answer D.**

The rule fails only on a card with an even number and a non-triangle. So check the even numbers, 4 and 10, in case a non-triangle is behind them, and the non-triangles, the circle and the square, in case an even number is behind them. The 7 cannot break the rule whatever is behind it, and nor can the triangle: the rule does not say that triangles only go with even numbers. Turning the triangle is the classic error of testing the converse.

**Question 5. Answer E.**

I: $n^3 - n = (n-1)n(n+1)$ is a product of three consecutive integers, so divisible by 2 and 3. II: $n^5 - n = n(n-1)(n+1)(n^2+1)$ contains $(n-1)n(n+1)$, so it is divisible by 6; for 5, check residues: if $n \equiv 0, \pm 1 \pmod 5$ one of the first three factors is a multiple of 5, and if $n \equiv \pm 2$ then $n^2 + 1 \equiv 5 \equiv 0$. III: $n^2 + n = n(n+1)$ is even, so $n^2 + n + 1$ is odd. IV fails for even $n$: $n = 2$ gives 3.



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**Question 6. Answer C.**

II, IV and V hold because $x^3$, $2^x$ and $x|x|$ are all increasing on the whole real line; $x|x|$ is $x^2$ for $x \geq 0$ and $-x^2$ for $x < 0$, and both pieces increase. I fails for $a = 1$, $b = -2$. III fails for $a = 1$, $b = -1$: then $\frac{1}{a} = 1$ is not less than $\frac{1}{b} = -1$. Taking reciprocals reverses an inequality only when both numbers have the same sign.

**Question 7. Answer F.**

On this interval $0 < \sin x < 1$ and $0 < \cos x < 1$. I: $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x > 1$ and the sum is positive. II: multiply $\sin x < 1$ by the positive number $\sin x$. IV: multiply $\cos x < 1$ by the positive number $\frac{1}{\cos x}$. III fails for any $x < 45°$.

**Question 8. Answer D.**

Take any two of the numbers, $p$ and $q$, and two others $r$ and $s$. Both $p + r + s$ and $q + r + s$ are even, so their difference $p - q$ is even: all five have the same parity. If all were odd, three of them would have an odd sum. So all five are even, and I, II and III follow. IV fails: $2, 2, 2, 2, 2$ has the property.

**Question 9. Answer C.**

Every pair consists of decreasing functions, so each is a counterexample exactly when $fg$ is not increasing. I: $e^{-2x}$ decreases, so it is a counterexample. II: $e^{2x}$ increases, so it is not. III: $x^2$ decreases for $x < 0$, so it is a counterexample. IV: $x^3e^x$ has derivative $x^2(x + 3)e^x$, negative for $x < -3$, so it is a counterexample. That makes three.

**Question 10. Answer A.**

Since $c$ is the longest side, $C$ is the largest angle, so the triangle is acute exactly when $C < 90°$, which by the cosine rule is exactly $a^2 + b^2 > c^2$. So I and II are sufficient, and so is III, since an equilateral triangle is acute. IV holds for every triangle with $a$ as the shortest side, including obtuse ones such as sides $3, 4, 6$, so it is not sufficient.

**Question 11. Answer D.**

C is true, so $N$ is one of $1, 4, 9, 16, 25$, and a perfect square above 1 is never prime, so A is false. We need exactly two of B, D, E. 1: none. 4: B only. 9: E only. 16: B and D. 25: D only. So $N = 16$. Option C, 12, makes exactly three statements true (B, D, E), but not with C among them.

**Question 12. Answer A.**

Every step holds. Line II divides by $a$, which is positive. Line IV is true, since $a + \frac{1}{a} - 2 = \frac{(a-1)^2}{a} \geq 0$. Line V combines a strict inequality with a non-strict one, $a + b > a + \frac{1}{a} \geq 2$, and the result is strict. The contradiction is genuine, so the proof is valid. Questions like this test whether you can confirm a correct argument rather than hunt for a mistake that is not there.

**Question 13. Answer D.**

Lines I to III are correct. In the $\sqrt{2}$ proof the corresponding step reads $2k^2 = q^2$, which forces $q^2$ to be even. Here the equation is $k^2 = q^2$, which says nothing about parity: indeed $p = 2$, $q = 1$, $k = 1$. Line IV is the first false step.



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**Question 14. Answer B.**

There are sign changes on $(0,1)$ and $(1,2)$, and since $f(x) \to \infty$ as $x \to \infty$ with $f(2) < 0$, a third root lies beyond 2: I is true. The roots are then $r_1 \in (0,1)$, $r_2 \in (1,2)$ and $r_3 > 2$, so $-b = r_1 + r_2 + r_3 > 3$ and II is true. IV is $d = f(0) < 0$. III can fail: with roots 0.5, 1.5 and 10 the local minimum is near $x = 7$.

**Question 15. Answer C.**

Take $f(x) = px^2 + qx + r$, so the integral is $\frac{2p}{3} + 2r$. I: $\frac{1}{3}\left((p - q + r) + 4r + (p + q + r)\right) = \frac{2p}{3} + 2r$. II: $2\left(\frac{p}{3} + r\right) = \frac{2p}{3} + 2r$. III gives $2r$, correct only when $p = 0$, and IV gives $2p + 2r$, also correct only when $p = 0$. So I and II. It is striking that two well-chosen points do as well as three.

**Question 16. Answer A.**

Statements 1 and 2 contradict each other, so exactly one of them is true, and statement 3 is false. If 1 were false, 2 would be true and 3 false, giving exactly one true statement, which would make 1 true: a contradiction. So 1 is true and 2 is false, and then exactly one statement is true, which is consistent. Only statement 1 is true.

**Question 17. Answer B.**

Line III is false as stated: for $x < 0$, $x + \frac{1}{x} \leq -2$, for instance $-2$ at $x = -1$. The correct fact is $\left|x + \frac{1}{x}\right| \geq 2$. That still rules out $t = 1$, and directly, $x^2 - x + 1 = 0$ has no real roots. So the conclusion stands: the polynomial is $(x-1)^2(x^2 - x + 1)$, whose only real root is 1. The answer is B.

**Question 18. Answer E.**

$f'(x) = 3x^2 + 2ax + b$. If $a^2 < 3b$, $f' > 0$ everywhere. If $a^2 = 3b$, $f'$ vanishes at one point and is positive elsewhere, and $f$ is still increasing, just as $x^3$ is. If $a^2 > 3b$, $f'$ is negative between its roots, so $f$ decreases there. The condition is $a^2 \leq 3b$. Option A demands $f' > 0$ everywhere, which is sufficient but not necessary.

**Question 19. Answer D.**

Two distinct positive roots means distinct real roots ($b^2 > 4c$), a positive product ($c > 0$) and a positive sum ($-b > 0$). With $c > 0$, the pair $b < 0$ and $b^2 > 4c$ is the single condition $b < -2\sqrt{c}$, so D is exactly right. E alone allows $c = 0$, which gives the roots 0 and $-b$. A allows complex roots, B allows two negative roots, C allows $c < 0$ with roots of opposite signs, and F allows a repeated root.

**Question 20. Answer E.**

Every $u_n$ is positive, so I holds. Squaring the recurrence gives $u_{n+1}^2 = u_n^2 + 4 + \frac{4}{u_n^2} > u_n^2 + 4$, and adding these from $n = 1$ gives II. II alone gives only $u_{100}^2 > 396$, not enough for III, so keep the error terms: $u_{100}^2 = 1 + 4 \cdot 99 + \sum_{n=1}^{99} \frac{4}{u_n^2} \geq 397 + \frac{4}{u_1^2} = 401 > 400$, and III holds. For IV, each $\frac{4}{u_n^2} \leq 4$, so $u_{100}^2 \leq 1 + 99 \cdot 8 = 793 < 900$ and IV is false. (In fact $u_{100} \approx 20.1$.)



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